Gentle Introduction to eBPF

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extended BPF

BPF: Berkeley / BSD Packet Filter

UC Berkeley

EFS (NFS) was constantly hitting 100% throughput utilization

No std way to export and analyse NFS metrics

possibilities and capabilities of eBPF programs

problem - - - - - - - - - - technology

Observability Gaps

Emerging Technology

kube-trace-nfs

https://github.com/4rivappa/kube-trace-nfs

Networking

l4 load balancers

facebook katran

cloudflare unimog

DDoS mitigation

cloudflare

other enterprise

container networking

cilium

calico

isovalent

Observability

beyla

netflix

hubble

other enterprise solutions

performance eng

Security

eBPF LSM

Falco

Tetragon

other enterprise solutions

in run-time

Android

network, battery

usage

Profilers

Many More ...

origin and history

cBPF, how eBPF evolved

programs and maps

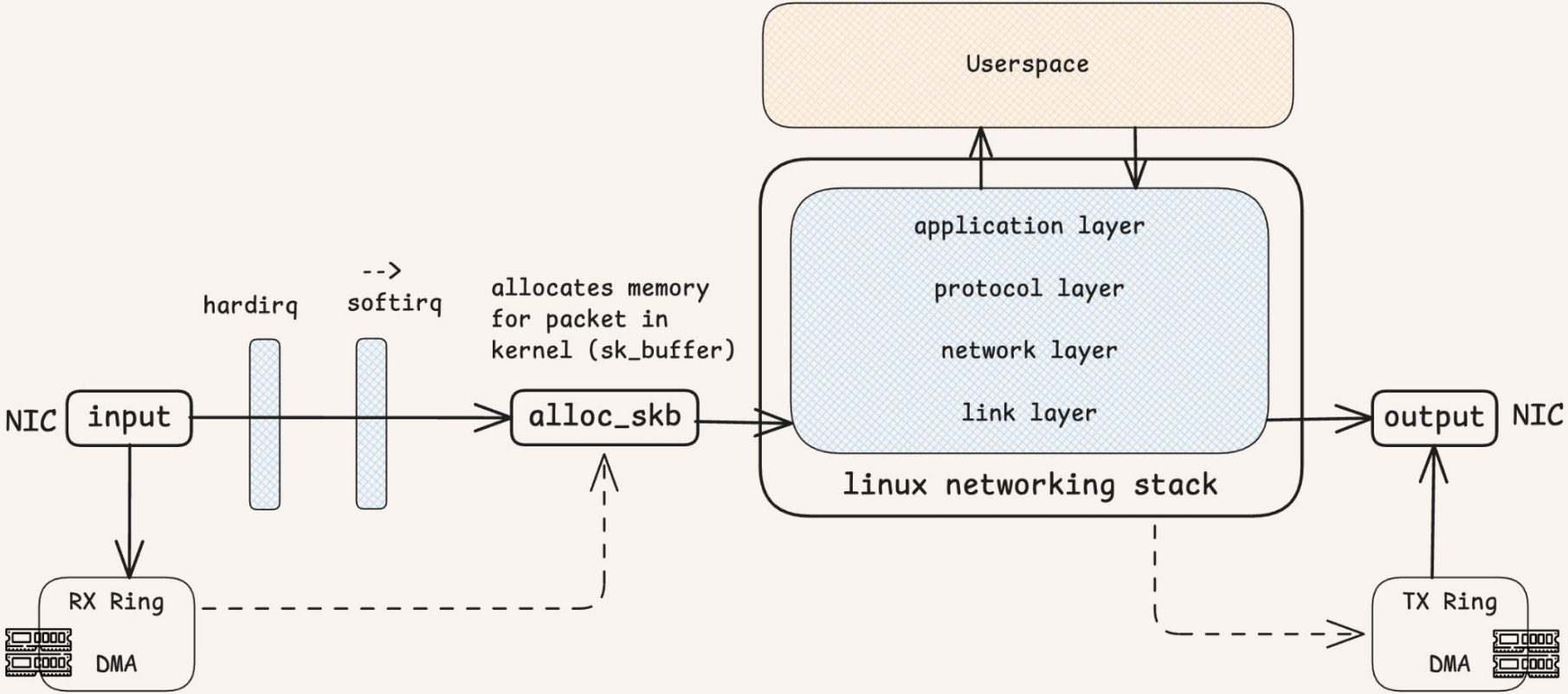
various types, use cases

load and attach

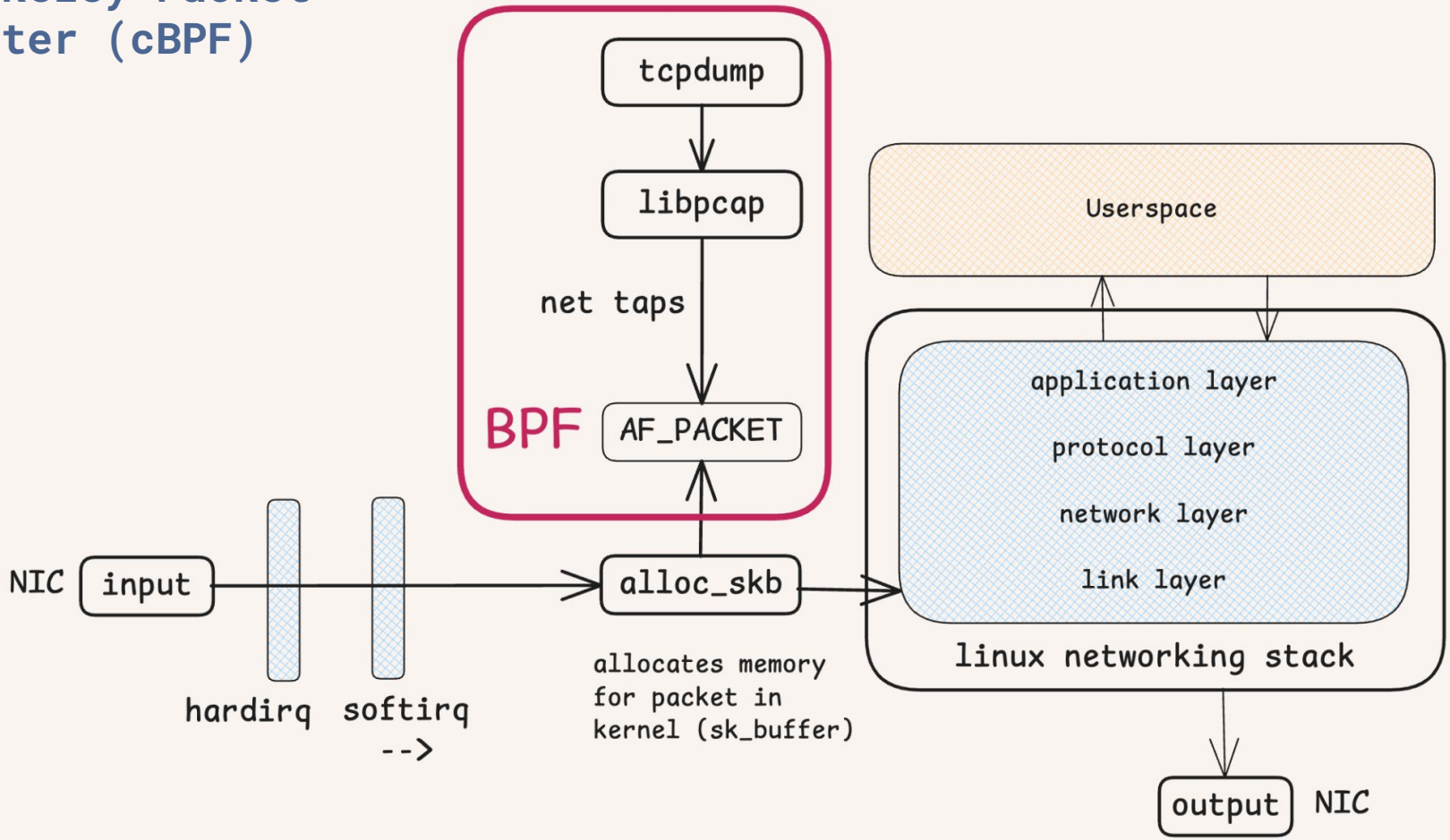
How prog’s are hooked

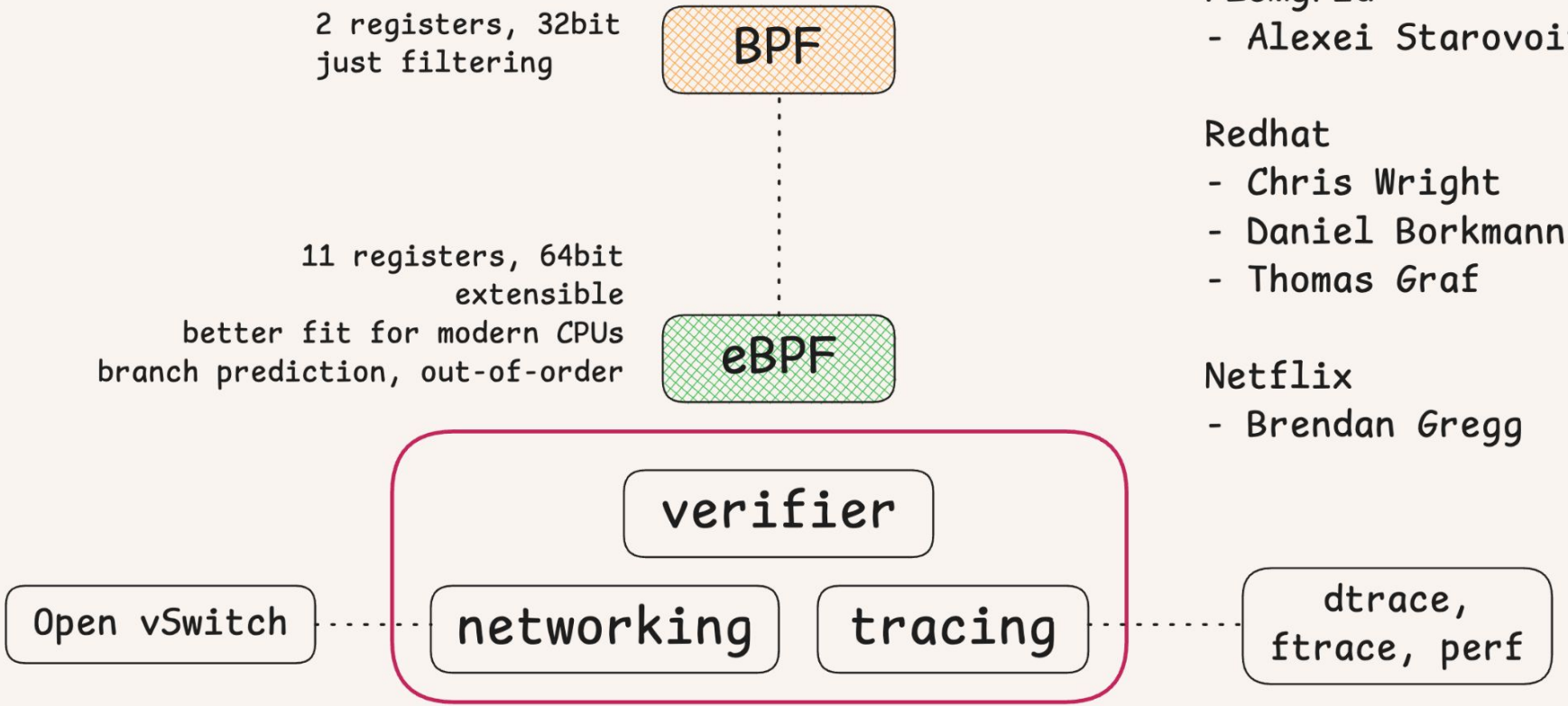
origin and history

Packet Journey



Berkeley Packet Filter (cBPF)





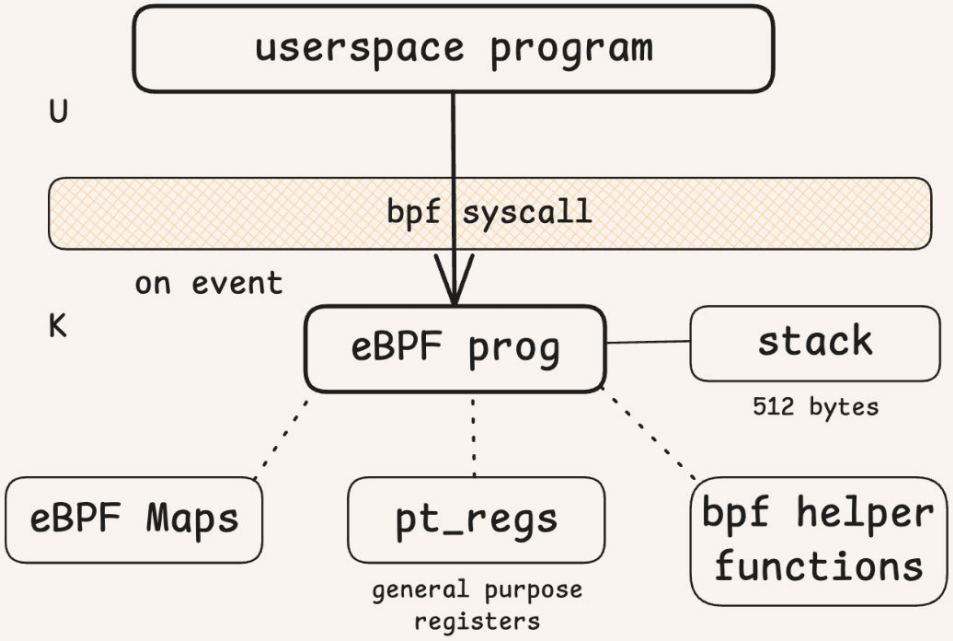
PLUMgrid
- Alexei Starovoitov

Redhat
- Chris Wright
- Daniel Borkmann
- Thomas Graf

Netflix
- Brendan Gregg

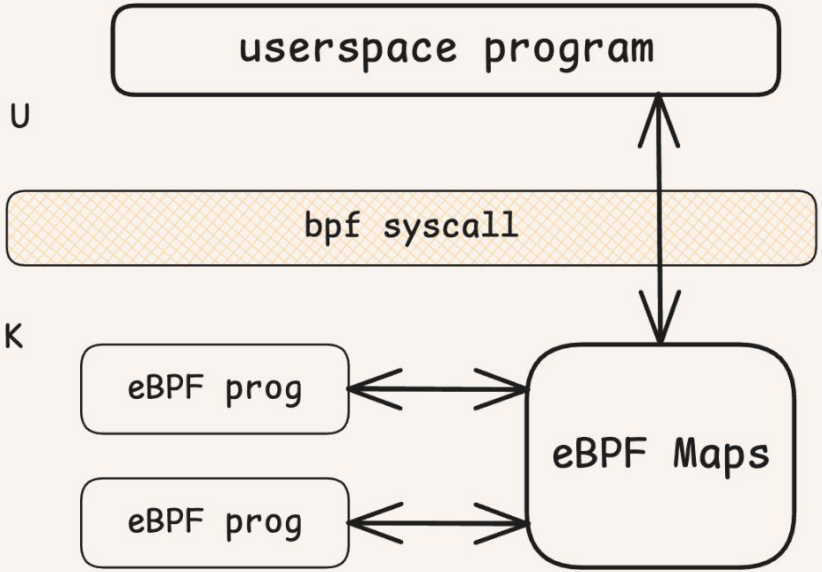
programs and maps

eBPF Programs



tracing	BPF_PROG_TYPE_TRACING BPF_PROG_TYPE_KPROBE BPF_PROG_TYPE_TRACEPOINT BPF_PROG_TYPE_PERF_EVENT
networking	BPF_PROG_TYPE_XDP BPF_PROG_TYPE_SCHED_CLS BPF_PROG_TYPE_SOCK_OPS BPF_PROG_TYPE_SK_SKB BPF_PROG_TYPE_SK_LOOKUP BPF_PROG_TYPE_NETFILTER BPF_PROG_TYPE_SOCKET_FILTER
other	BPF_PROG_TYPE_CGROUP_SKB BPF_PROG_TYPE_CGROUP_SOCK BPF_PROG_TYPE_CGROUP_SYSCTL
	etc ...

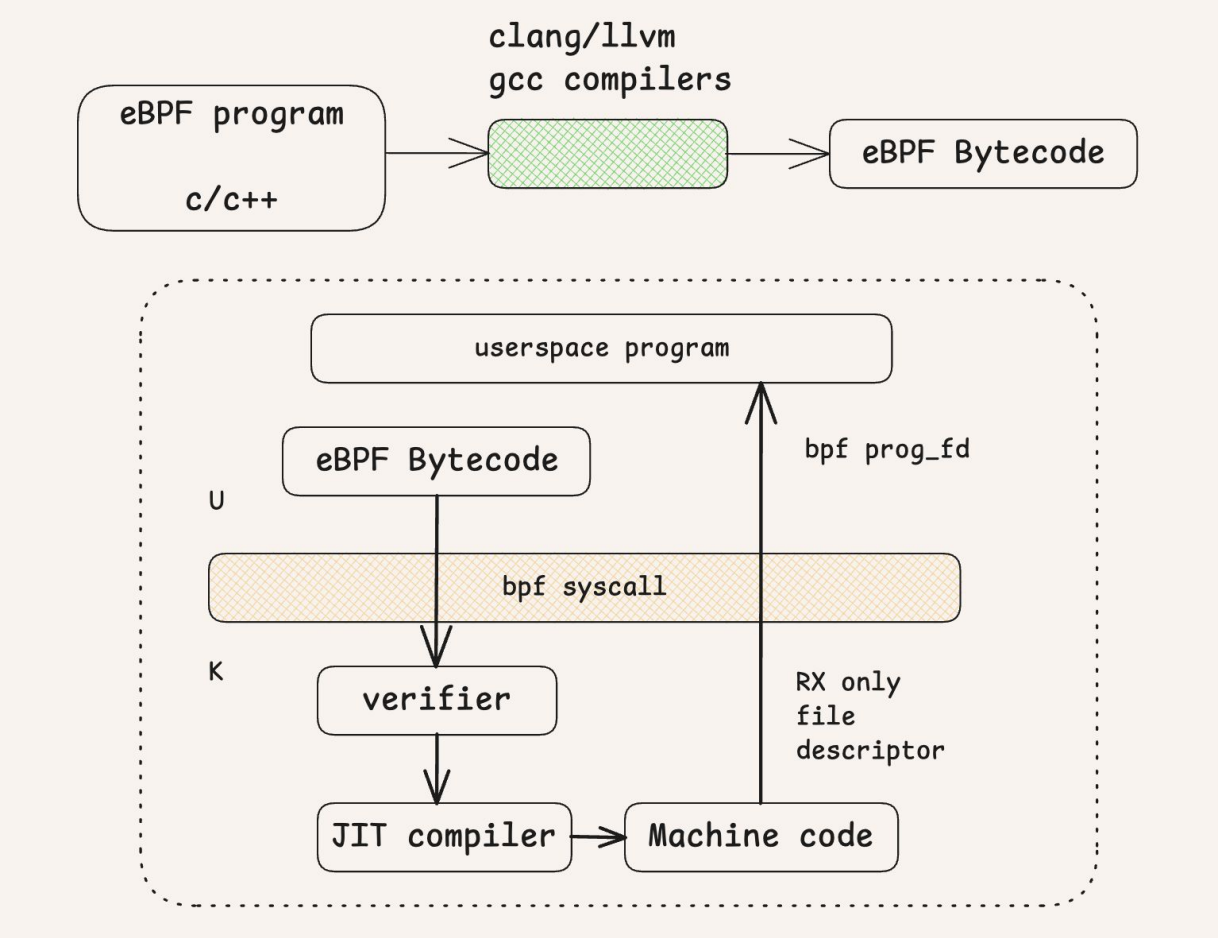
eBPF Maps



generic	BPF_MAP_TYPE_HASH BPF_MAP_TYPE_ARRAY BPF_MAP_TYPE_QUEUE BPF_MAP_TYPE_STACK
ebpf maps	BPF_MAP_TYPE_ARRAY_OF_MAPS BPF_MAP_TYPE_HASH_OF_MAPS
other	BPF_MAP_TYPE_CGROUP_STORAGE BPF_MAP_TYPE_CGRP_STORAGE BPF_MAP_TYPE_BLOOM_FILTER BPF_MAP_TYPE_CPUMAP BPF_MAP_TYPE_DEVMAP BPF_MAP_TYPE_LPM_TRIE BPF_MAP_TYPE_SK_STORAGE BPF_MAP_TYPE_SOCKMAP etc ...

load and attach

prog load

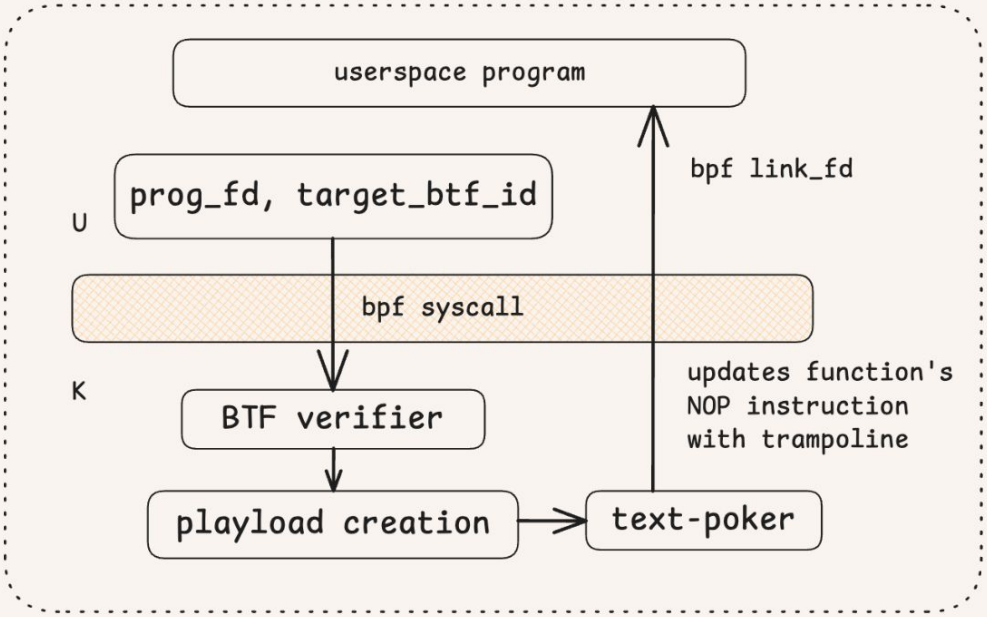

clang/llvm
gcc compilers
eBPF program
c/c++
eBPF Bytecode
userspace program
eBPF Bytecode
bpf prog_fd
U
bpf syscall
K
verifier
RX only
file
descriptor
JIT compiler
Machine code

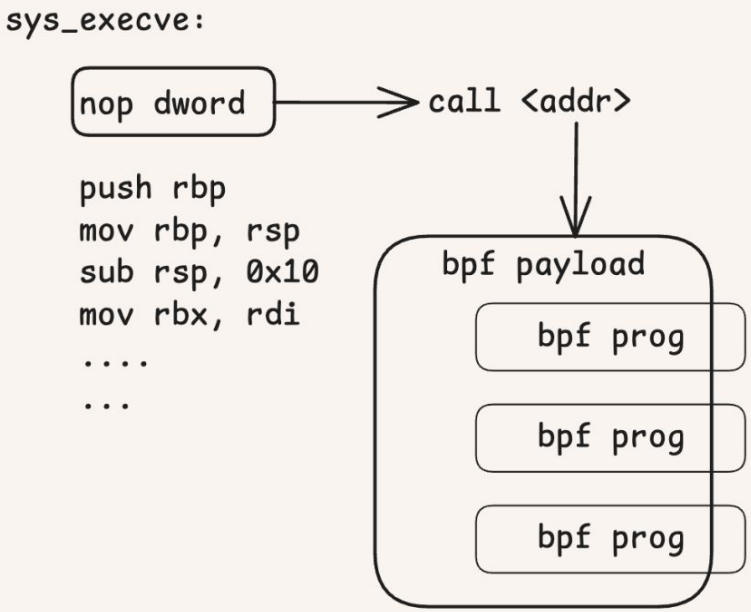
prog attach

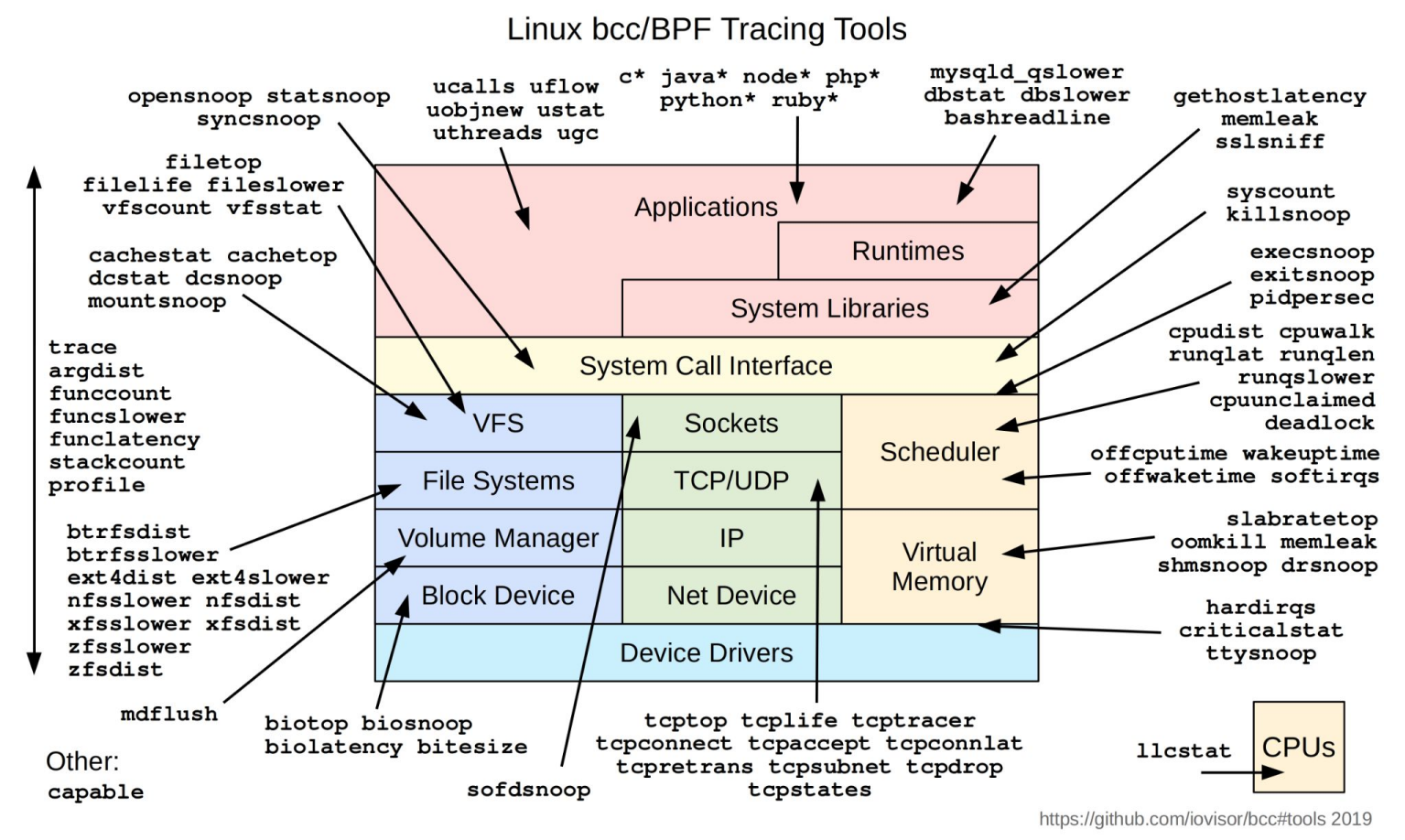
```
/proc/kallsyms

ffff8000806a8c98 T __arm64_sys_execve

[75941] FUNC '__arm64_sys_execve' type_id=75774
```







source: github.com/iovisor/bcc#tools (Brendan Gregg)

- BCC tools (helpful scripts to analyze majority of system events)
  - https://github.com/iovisor/bcc

- bpftrace (single liners and short scripts to debug, live systems)
  - https://github.com/bpftrace/bpftrace
  - checking all attach points on your system: “bpftrace -l”

- bpftool (managing bpf programs and maps on systems)
  - https://github.com/libbpf/bpftool

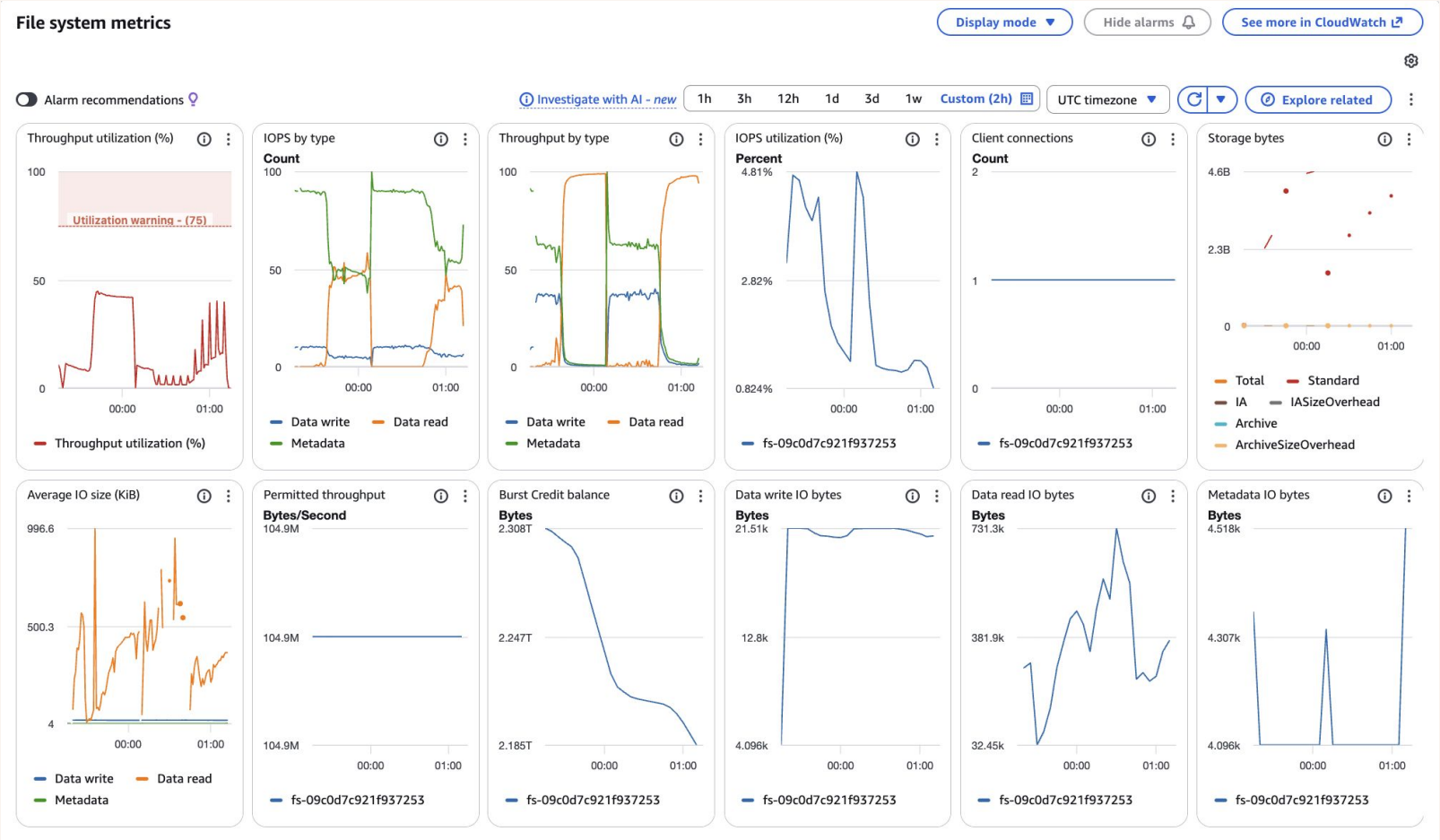
- beyla (eBPF-based auto instrumentation for applications)
  - https://grafana.com/oss/beyla-ebpf/

- inspektor-gadget (observability on kubernetes clusters using eBPF)
  - https://github.com/inspektor-gadget/inspektor-gadget

- eBPF libraries
  - c/c++ (libbpf)
  - go (ebpf, libbpfgo)
  - rust (aya, libbpf-rs)

TODO: Demo NFS Flamegraph

AWS EFS Metrics (NFS)

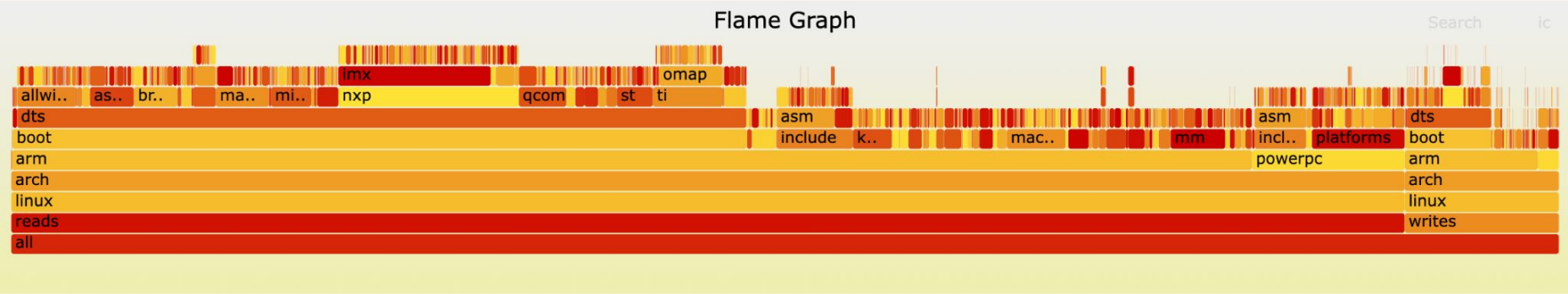


kube-trace-nfs

- https://github.com/4rivappa/kube-trace-nfs

nfs-flamegraph

- https://github.com/4rivappa/nfs-flamegraph



Beyla eBPF (OpenTelemetry eBPF Instrumentation)

- https://grafana.com/oss/beyla-ebpf/
- https://opentelemetry.io/docs/zero-code/obi/

- Zero Code Auto-Instrumentation
- Key metrics of requests (throughput, errors, latencies)
- Distributed tracing and context
- Network observability (flows btw services)

References

- eBPF documentary: https://youtu.be/Wb_vD3XZYOA
- Initial patch commit of eBPF: https://github.com/torvalds/linux/commit/bd4cf0ed331a275e9bf5a49e6d0fd55dffc551b8
- cBPF: https://www.tcpdump.org/papers/bpf-usenix93.pdf
- Linux networking stack: https://commons.wikimedia.org/wiki/File:Netfilter-packet-flow.svg
- L4 load balancers:
  - https://blog.cloudflare.com/unimog-cloudflares-edge-load-balancer/
  - maglev:https://static.googleusercontent.com/media/research.google.com/en//pubs/archive/44824.pdf
  - https://github.com/facebookincubator/katran
- Networking:
  - cilium: https://github.com/cilium/cilium
  - https://github.com/cilium/tetragon
  - https://github.com/microsoft/retina